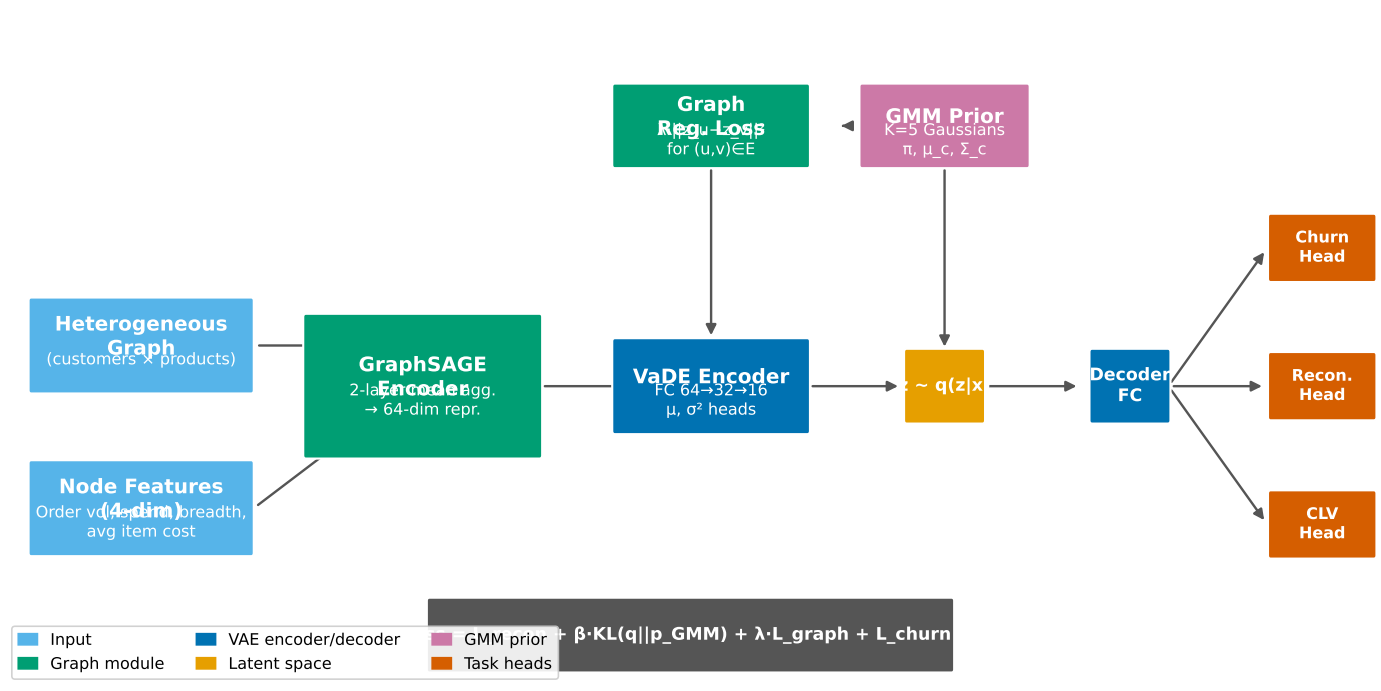
Heterogeneous Graph
(customers × products)
Node Features
(4-dim)
Order velocity, breadth, avg item cost
GraphSAGE Encoder
2-layer mean-agg.
→ 64-dim repr.
Graph Reg. Loss
for (u,v)∈E
GMM Prior
K=5 Gaussians
π, μ_c, Σ_c
VaDE Encoder
FC 64→32→16
μ, σ² heads
z ~ q(z|x)
Decoder FC
Churn Head
Recon. Head
CLV Head
+ β·KL(q||p_GMM) + λ·L_graph + L_churn
Input
Graph module
VAE encoder/decoder
Latent space
GMM prior
Task heads